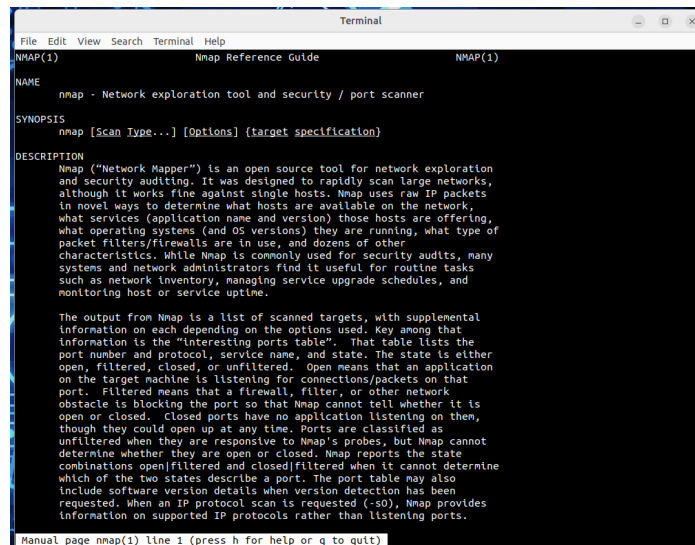


REPORT S11L5

Esercizio 3: Esplorazione di Nmap



Cos'è Nmap? Per cosa viene usato nmap?

Nmap è un software open-source il più usato, la sua utilità è scansionare la rete: gli indirizzi IP, le porte aperte o chiuse appartenenti a ciascuno, i protocolli e anomalie alle versioni ad un eventuale aggiornamento vecchio o una possibile backdoor, quindi per farla in breve scansiona le vulnerabilità.

```
[analyst@secOps ~]$ nmap -A -T4 Scanme.Nmap.Org
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-19 11:41 UTC
Nmap scan report for Scanme.Nmap.Org (45.33.32.156)
Host is up (0.17s latency).
Other addresses for Scanme.Nmap.Org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
rDNS record for 45.33.32.156: scanme.nmap.org
Not shown: 996 closed ports
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Go ahead and ScanMe!
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 17.65 seconds
[analyst@secOps ~]$
```

Qual è il comando nmap usato? Cosa fa l'opzione -A? Cosa fa l'opzione -T4?

E' stato utilizzato il comando -A -T4.

-A serve ad avviare una scansione aggressiva del target Scanme.Nmap.org
-T4 invece è in modalità aggressivo e accelera i timeout e riduce l'attesa dei pacchetti.

```
[analyst@secOps ~]$ nmap -A -T4 localhost
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-19 11:59 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 127.0.0.1
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_At session startup, client count was 2
|_vsFTPD 3.0.5 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.1 (Ubuntu Linux; protocol 2.0)
|_ssh-hostkey:
|_256 be:7d:82:5e:38:db:f3:54:a3:cd:75:b2:ff:b5:ce:29 (ECDSA)
|_256 12:e2:5b:5c:ef:f1:35:eb:9e:8f:a2:2f:39:de:94:18 (ED25519)
23/tcp    open  telnet   Linux telnetd
Service Info: Host: Welcome; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submitt/.
Nmap done: 1 IP address (1 host up) scanned in 18.62 seconds
[analyst@secOps ~]$
```

Quali porte e servizi sono aperti?

Scansionato il mio localhost con gli stessi comandi del precedente vediamo che abbiamo 1 host attivo 3 porte aperte

21/tcp al servizio ftp con la versione vsftpd 2.0.8

22/tcp al servizio ssh e versione OpenSSH 8.9p1 Ubuntu

23/tcp al servizio telnet versione Linux telnetd

```
[analyst@secOps ~]$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:44:07 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.21/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 83113sec preferred_lft 83113sec
    inet6 fe80::a00:27ff:fe55:4407/64 scope link
        valid_lft forever preferred_lft forever
3: ovs-system: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 16:fc:f3:5f:3c:a0 brd ff:ff:ff:ff:ff:ff
4: s1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 6a:45:e2:1d:2e:41 brd ff:ff:ff:ff:ff:ff
[analyst@secOps ~]$
```

A quale rete appartiene la tua VM?

La mia rete della VM appartiene a 192.168.1.21/24

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ nmap -A -T4 192.168.1.21/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-19 12:15 UTC
Nmap scan report for home (192.168.1.1)
Host is up (0.019s latency).
Not shown: 988 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      0.0.0
fingerprint-strings:
  Genericlines, NULL, SMBProgNeg, SSLSessionReq:
  220----- Welcome to FTP Server -----
  220-You are user number 1 of 50 allowed.
  220-Local time is now 13:15. Server port: 21.
  220-IPv6 connections are also welcome on this server.
  will be disconnected after 15 minutes of inactivity.
  Help:
  220----- Welcome to FTP Server -----
  220-You are user number 1 of 50 allowed.
  220-Local time is now 13:15. Server port: 21.
  220-IPv6 connections are also welcome on this server.
  will be disconnected after 15 minutes of inactivity.
  following SITE commands are recognized
  CHMOD
  IDLE
  UTIME
22/tcp    filtered ssh
23/tcp    open  telnet   0.0.0
fingerprint-strings:
  Genericlines:
  DX3301-T1 Username:
  DX3301-T1 Username: DX3301-T1 Username:
  GetRequest:
  HTTP/1.0
  DX3301-T1 Username:
  Help:
  HELP
  DX3301-T1 Username:
  NCP:
  DmdT^@^@^@
  ^@^@^@A^@^@^@
  NULL:
  DX3301-T1 Username:
  SIPOptions:
  OPTIONS sip:nm SIP/2.0
  VLa: SIP/2.0/TCP nm;branch=foo
  From: <slp:nm@nm>;tag=root
  <slp:nm2@nm2>
  Call-ID: 50000
  CSeq: 42 OPTIONS
```

Quanti host sono attivi?

```
Terminal
File Edit View Search Terminal Help
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (4 hosts up) scanned in 196.45 seconds
[analyst@secOps ~]$
```

Abbiamo esattamente 4 hosts attivi

```

[analyst@secOps ~]$ nmap -A -T4 scanme.nmap.org
Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-19 12:31 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.17s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
80/tcp    open  http
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Go ahead and ScanMe!
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.24 seconds
[analyst@secOps ~]$

```

Quali porte e servizi sono aperti?

Qui abbiamo le porte e i servizi aperti:

```

22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped

```

Quali porte e servizi sono filtrati?

Nessun filtrato ma risulta 996 closed ports

Qual è l'indirizzo IP del server?

45.33.32.156

Qual è il sistema operativo?

Il sistema operativo è un Linux Ubuntu

Domanda di Riflessione Nmap è uno strumento potente per l'esplorazione e la gestione della rete. Come può Nmap aiutare con la sicurezza della rete? Come può Nmap essere usato da un attore malevolo come strumento nefasto?

Grazie a Nmap ti permette di analizzare la rete in modo da scoprire le possibili vulnerabilità e intervenire nel modo più rapido.